

Azure AI Voice Live API vs Azure OpenAI GPT Realtime API (GA deep dive)

Bottom line: two-GA-API deep dive. The repo also includes a Foundry voice agent public preview; see the root README for the full three-way table. Same model family, same browser app, same Python bridge pattern; the difference is who manages deployment, capacity/quota, voice/audio processing, and lifecycle risk.

Use Voice Live when...

- You want managed models and no model deployment.
- You prefer per-resource Voice Live limits over per-model deployment quota.
- You need Azure neural/HD/custom voices, built-in transcription, semantic VAD, noise suppression, echo cancellation, Avatar, BYOM, or agent mode.
- You can request new-connections/min increases when concurrency requires it.

Use Realtime when...

- You need model version, deployment name, capacity, and upgrade control.
- You already have realtime quota and regional capacity.
- You need the GA nested schema, WebRTC ephemeral tokens, or direct SIP on the realtime surface.
- You accept no PTU, constrained regions, quota forms, and lifecycle-date management.

Third option: Foundry voice agent is public preview. It stores instructions, tools, voice, greeting, audio settings, versions, traces, and transcripts in a Foundry project; the bridge sends no session config and executes the tools (shared RAG). It connects through the project route `/api/projects/<p>/agents/<a>/endpoint/protocols/voice` (the older Voice Live agent-mode route disconnects kind: voice agents), runs on managed models with no deployment, and shares Voice Live limits with the Voice Live app in shared-platform mode. Do not position it as production-ready.

Core comparison

Dimension	Voice Live API	Realtime API
Status / endpoint	GA 2026-07-15; wss://<foundry>.services.ai.azure.com/voice-live/realtime?api-version=...&model=...	GA /openai/v1; wss://<resource>.openai.azure.com/openai/v1/realtime?model=<deployment>; no api-version.
Provisioning	AIServices S0 resource, custom subdomain, local auth disabled; no model deployment.	AIServices plus accounts/deployments Global Standard model deployment with capacity and version policy.
Models	gpt-realtime-mini GA Basic default; gpt-realtime-2.1-mini preview Basic.[4]	gpt-realtime-2.1-mini 2026-07-07 GA default, retires 2027-06-25; gpt-realtime-mini retirement rows conflict.[1]
Capacity	Not a deployment. Per resource: ≤120K TPM, 100 new connections/min, ≤60-min sessions; a support request raises connections/min, and TPM follows (TPM = NCPM × 4,000).	Deployment capacity units : gpt-realtime-2.1-mini 1 unit = 10,000 TPM + 20 RPM (default quota often 10 units = 100K TPM / 200 RPM; the portal shows TPM). Pooled per subscription + model version; no PTU.[3]
Regions	centralus, eastus2, swedencentral, westus2 confirmed for demo default; coverage varies.	Global Standard: canadacentral, centralus, eastus2, francecentral, swedencentral, southindia; DZS conflict.[5]
Auth / RBAC	Scope https://ai.azure.com/.default; Cognitive Services User a97b...908 + Foundry User 53ca...56d.	Scope https://ai.azure.com/.default; Cognitive Services OpenAI User 5e0b...61bd.
Schema / events	Flat session.update; response.audio.delta, response.audio_transcript.delta, conversation.item.created.	Nested GA schema; response.output_audio.delta, response.output_audio_transcript.delta, conversation.item.added/.done, rate_limits.updated.
Audio features	Azure/OpenAI voices, Azure semantic VAD, deep noise suppression, server echo cancellation, built-in transcription.	OpenAI voices, semantic/server VAD, near/far-field noise reduction, optional separate transcription deployment.
Transports / extras	Server WSS in demo, with shared ACS, Twilio, and Asterisk (chan_websocket) adapters; Avatar, BYOM, agent mode, timestamps/visemes.	Server WSS in demo with the same ACS/Twilio/Asterisk adapters; WebRTC client secrets, /realtime/calls, SIP in limited regions.
Pricing / monitoring	Per-1M-token Voice Live tiers; mini = Basic; no quoted dollar prices.[2] App metrics to Log Analytics.	2.1-mini Global per 1M: text \$0.60/\$0.06 cached/\$2.40 out; audio \$10/\$0.30 cached/\$20 out. Add Azure OpenAI request/token/seconds metrics.

Code changes

208 lines Voice Live bridge vs **193 lines** Realtime bridge (**224 lines** preview voice-agent bridge).

Hook	Delta
build_url	Voice Live adds /voice-live/realtime, api-version, and model query. Realtime adds /openai/v1/realtime and deployment query.
build_headers	Both use Entra; Realtime scope is env-overrideable.
session.update	Voice Live flat schema + Azure audio features; Realtime nested GA schema + optional transcription deployment.
Events	Shared bridge accepts both audio/item naming families.
Infra	Realtime adds deployment resource, model version map, capacity param (with a pre-provision quota check), and OpenAI User role. Every example creates a Foundry resource with a voice-agents project; everything else is shared.

Capacity implication

Formula: p90 TPM/call x concurrent calls. Example: ~80K TPM/call x 20 calls ~ = 1.6M TPM. Both APIs need sizing; Voice Live and the voice agent shift from deployment quota to per-resource connection/TPM limits; in shared mode they share the same limit.

Reduce tokens: shorter instructions/tools, concise spoken answers, history cap/delete, barge-in truncation, and realistic load tests.

Confirm before cutover: approved quota/limit increase, validated region capacity, p90 TTFA, busy count, and rollback path.

Migration path and reproduction

- Change endpoint, auth scope if needed, and RBAC.
- Remove or add model deployment depending on direction.
- Reshape session.update flat ↔ nested.
- Map events; shared bridge already accepts both.
- Pick voice/VAD/noise/transcription options.
- Load test; request Voice Live connection increase or Realtime quota before cutover.

First pin the tenant: az login --tenant <TENANT_ID>, azd auth login --tenant-id <TENANT_ID>, then per example azd env set AZURE_TENANT_ID / AZURE_SUBSCRIPTION_ID / AZURE_LOCATION. Never a bare azd up.

Deploy: cd examples\voice-live-api, cd examples\realtime-api, or cd examples\foundry-voice-agent, then azd up. For the voice agent, profile edits require azd hooks run postprovision.

See 03-deployment.md and 04-testing.md. Keep both behind the same bridge interface until cost, quality, and capacity pass.

Sources

Verified 2026-09-25; updated 2026-09-29 (capacity units from the live deployment and model catalog, voice-agent project route, Asterisk telephony). Sources: Voice Live API reference, overview, release notes, quotas, regions, and pricing pages; Azure OpenAI Realtime how-to, WebRTC how-to, model retirement schedule, quotas/limits, quota management, region availability, pricing page, and Azure Retail Prices API. Repo code inspected: both bridges, both resources.bicep, both azure.yaml, both READMEs, and shared bridge. Re-verify model lifecycle, regions, quota, and pricing before customer use. Footnotes: [1] retirement conflict; [2] no Voice Live dollars quoted; [3] capacity units observed 2026-09-29 via az cognitiveservices model list and a live deployment (the usage row is labelled Requests Per Minute but counts units); [4] Voice Live 2.1-mini preview; [5] DZS availability conflict.